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# Heater Pigging & Decoking Services

## Valero Texas City Refinery

Mechanical pigging & decoking for fired heaters

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# Agenda

Heater pigging & decoking at Texas City

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01 Pigging & decoking methodology

02 Execution

03 Readiness & resources

04 Equipment & technology

05 Cleanliness verification & QA/QC

06 Emergency response

07 Safety & risk mitigation

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10 Why USA DeBusk

SERVICES

# Our services for Texas City

Mechanical decoking of fired heater coils, planned or on short notice

### Coker heaters

Hard coke in radiant and convection coils, on run-length or turnaround cycles

### Crude & vacuum heaters

Pitch, resid and coke across mixed-diameter passes

### Charge & process heaters

Process heater coils cleaned and verified ahead of restart or inspection

### Smart pig support

Coils cleaned to inspection standard; spare TriMax pump drives the inspection tool

**Planned**

turnarounds & run-length decokes

**Short notice**

emergency mobilization

**Optional**

closed-loop filtration



01 METHODOLOGY

# Bi-directional mechanical decoking

Every pass follows the same controlled sequence

1. **Rig-in & pressure test**
   Connect 300# launchers and receivers; fill and test before any launch
2. **Baseline flow test**
   Record pressure at a set rate for a before-and-after comparison
3. **Gauge foam run**
   Proves the pass is open and sets the initial pig size
4. **Scraper pig progression**
   Carbide-studded pigs run in both directions, stepped up ⅛" per clean pass
5. **Oversize polish**
   Up to tube ID + ¼" maximum, with real-time pressure monitoring
6. **Gauge foam & sign-off**
   Line-size foam witnessed by Valero, final flow test, pass signed clean




*Scraper pigs after runs: wear marks show where coke remains*

# Pig progression plan

The gauge foam run sets the starting size; the smallest tube ID sets the ceiling

## Example: 4" Sch 80 coil

Tube ID 3.826"




## Example: mixed-diameter pass, 3" to 8"

Smallest ID 2.900" governs the whole pass




### The sizing rule

**Max pig OD = smallest tube ID + 0.250"**

- Step up ⅛" after each clean pass
- Pig wear drives size & appendages
- Body length set for bends & mule ears
- Gauge foam sets the starting size
- Line-size foam witnessed at sign-off

01 METHODOLOGY

# Engineered decoking pigs, matched to the coke

Body, durometer and studding chosen for the coil, the coke and the metallurgy



*New engineered decoking pigs*

### Soft-core and hard-core designs

Urethane or high-density foam bodies, plus foam pushers for flushing and proving the pass

### A full range of durometers

Soft to hard compounds, chosen to compress through tight bends or cut hard coke

### Bald or studded

Tungsten carbide, hardened-steel and stainless stud options, matched to the tube metallurgy

### Heavy-coke designs

High-durometer, fully studded pigs brought in when pig wear shows hard, heavy coke

01 METHODOLOGY

# Coil-specific controls

Every heater is planned from its drawings before we mobilize

### Tube geometry

Footage, tube IDs and diameter changes per pass set the pig progression and the duration

### Metallurgy

Stainless and chrome coils: water quality verified before fill, no galvanized contact with tubes

### Connections

Coil flange size and rating confirmed; spools and adapters matched to our 300# launchers and receivers

### Return bends

Pig length and flexibility selected for 180° returns, plug headers and mule ears

### Launcher access

Inlet and outlet elevations walked down; rig layout set before rig-in

### Coke & service

Coker, crude or vacuum service decides which pigs are staged on day one

02 EXECUTION

# How a decoke runs

Continuous 24/7 pigging on 12-hour shifts; passes pigged in conjunction

**Plan**

Drawings reviewed, pig progression and rig layout set

**Rig-in**

Launchers and receivers connected, filled and pressure tested

**Pig**

Up to three passes at once per TriMax, around the clock

**Rig-over**

Move to the next set of passes, if the heater has more

**Verify**

Final gauge foam and flow test, pass signed off with Valero

**Rig-out**

Equipment out, job book delivered

## 24/7

two 12-hour shifts until the heater is handed back

## Up to 3

passes pigged at once on each TriMax

## 2+

TriMax units on larger heaters or multi-heater jobs

# Crew & equipment, every job

## Equipment

| | |
|---|---|
| 1 | TriMax triple pumper |
| 1 | Support unit: fittings, tooling, spares |
| 2 | Crew trucks |

## Manpower

| | |
|---|---|
| 2 | Supervisors: 1 day, 1 night |
| 3 | Operators: 2 day, 1 night |
| 1 | Project manager |




**5%**
approximate crew turnover, last 12 months

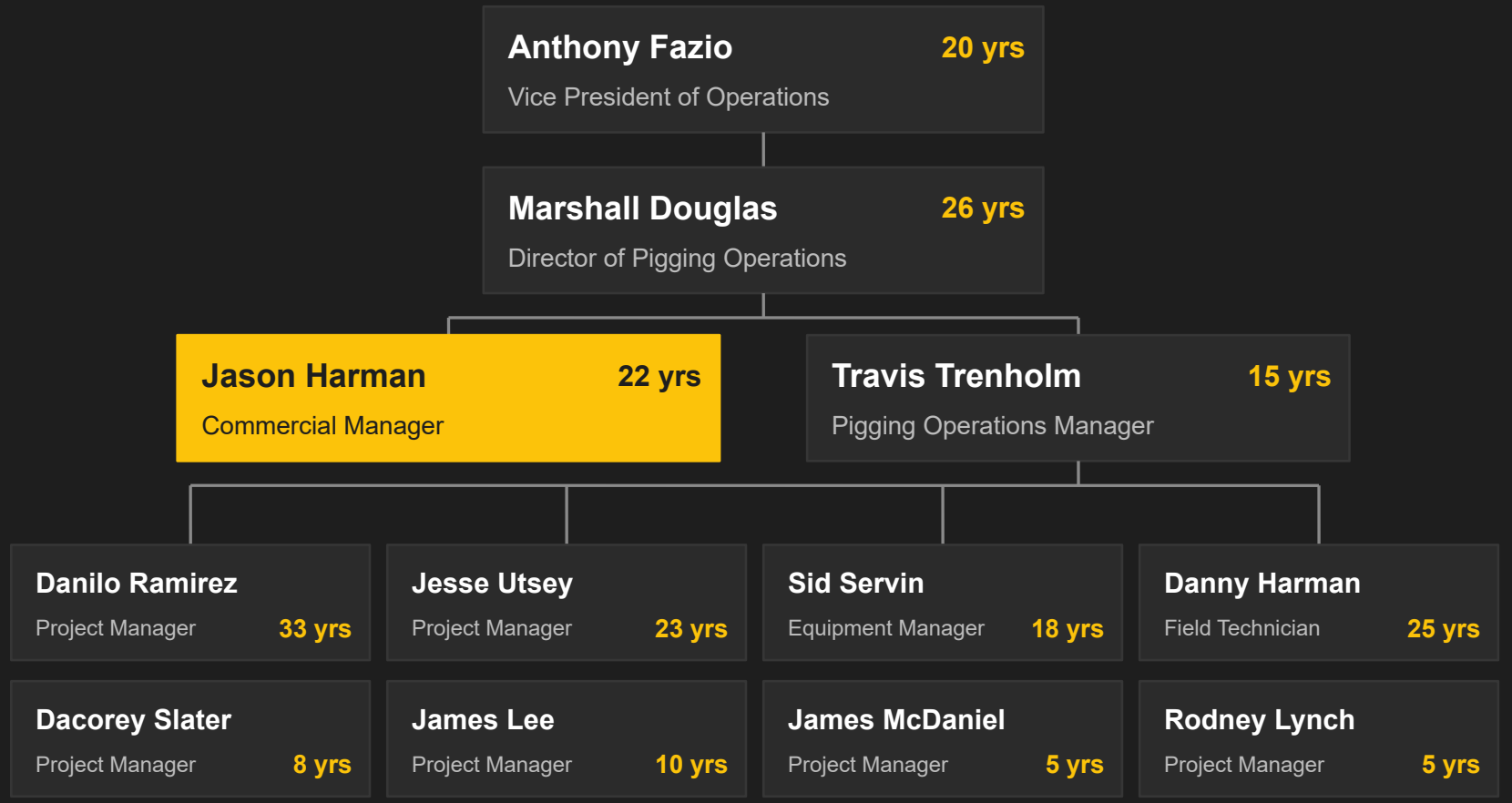
**24/7**
two 12-hour shifts

**100%**
training verified before mobilization

# Our pigging team

**210**
years of combined pigging & decoking experience

**15+**
years self-performing heater pigging



# The TriMax triple pumper

6 identical TriMax units, 18 engines: one standardized fleet

### Three independent pump systems

Closed-loop, bi-directional; 50% more productive than a twin pumper

### Automated HMI control

Three 17" touchscreens, automated digital pig logs

### 150–300 psi working pressure

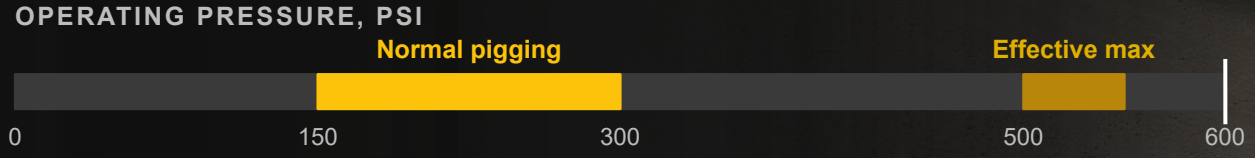
Pump rated 600 psi; 500–550 psi effective after valve losses

### Real-time sensor data

Coke shows as pressure spikes at the operator interface

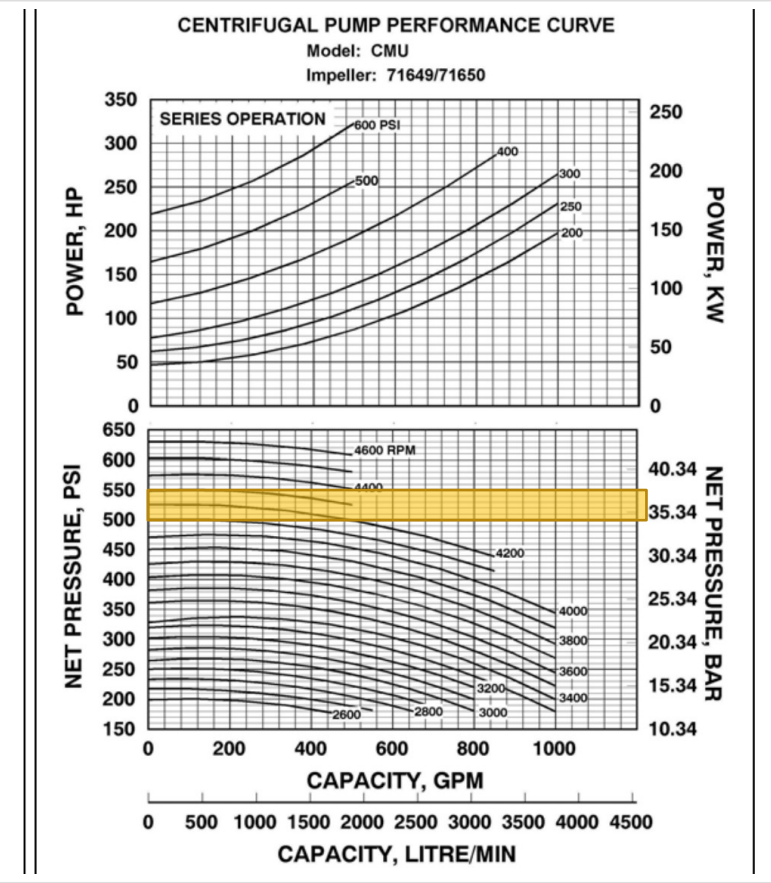
### Cummins QSL9, 333 bhp, Tier 4 Final

The same engine across all 18 pumps in the fleet



# TriMax pump performance

Waterous CMU two-stage pump, one per pass: series-operation curve, as used for pigging




**500–550 psi**
Effective maximum at the pig, after losses through our valves

**Reading the curve** Each line is one engine speed; pressure falls as flow rises. Pigging runs well inside this envelope at 150 to 300 psi.

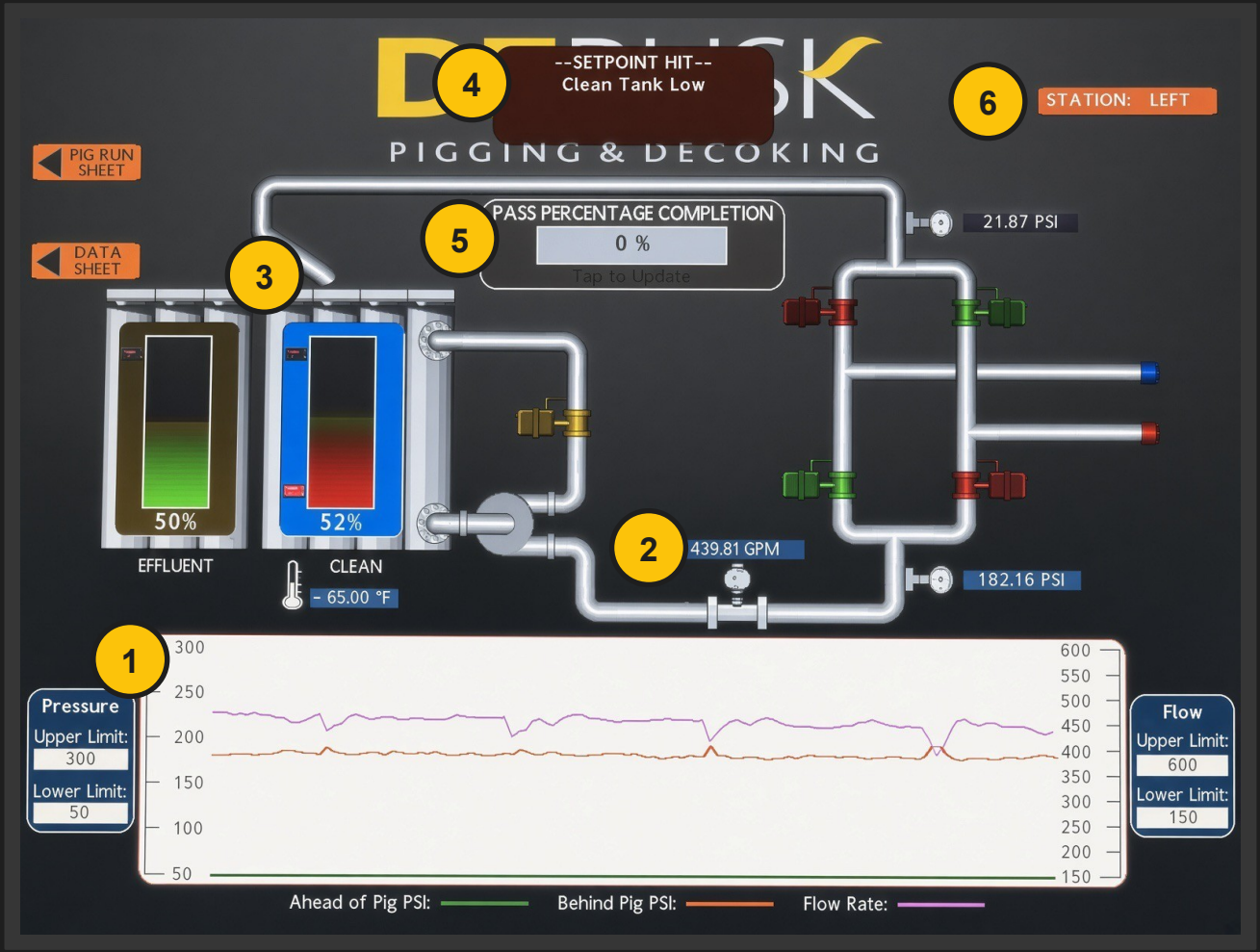
**150–300**
psi normal pigging

**500–550**
psi effective max, after valve losses

**600**
psi rated pump maximum

*Manufacturer pump curve: Waterous Company*

# Real-time control at every operator station




1. **Live trends**
   Pressure ahead of and behind the pig, plus flow; coke shows as a spike
2. **Flow & pressure**
   GPM and psi at the pump, updated continuously
3. **Tank levels**
   Clean and effluent tanks on guided-radar indicators
4. **Setpoint alarms**
   Upper and lower limits on pressure and flow trigger alerts
5. **Digital pig logs**
   Pass completion, pig run sheet and data sheet recorded automatically
6. **One station per pass**
   Left, center and right stations run independently

# Optional closed-loop filtration

Runs alongside pigging, independent of coil pressure




*Filtrate returns to the clean tank; pigging continues without interruption*

**400 GPM**

press capacity

**1,243 ft²**

filtration area per press

**3**

trailer-mounted presses in the fleet

*Optional: priced separately*

- **Reduces fresh-water demand**
- **Reduces wastewater volume**
- **Coke fines captured as filter cake for disposal**

05 CLEANLINESS VERIFICATION & QA/QC

# Verifying every pass is clean

Four independent indicators, reviewed side by side with Valero

### 1 Sensor data

Coke shows as a pressure spike; the operator works that location until it clears

### 2 Effluent returns

Coke fines in the return stream taper off to nothing as the pass cleans

**Joint sign-off per pass**



### 3 Pig condition

Wear marks guide pig size and appendage choices

### 4 Final gauge foam

Soft line-size (or slightly over) foam, witnessed by the Valero rep

*A pass is called clean only when all four indicators agree.*

05 CLEANLINESS VERIFICATION & QA/QC

# QA/QC from walkdown to job book

Governed by the USA DeBusk QA/QC manual

## Before

- Pre-job walkdown with operations
- Coil data checked against Valero records
- JSA, permits and isolation confirmed
- Pig progression staged by heater

## During

- Daily report to one point of contact
- Automated digital pig log per pass
- Flow tests before and after at the same rate
- JSA reviewed at every shift change

## Close-out

- Per-pass cleanliness sign-off
- Final pig size recorded per pass
- Job book delivered to Valero
- Post-job review and lessons learned

06 EMERGENCY RESPONSE

Based in Deer Park, TX

# Responsive support for Texas City

Our fleet stages at our Deer Park base, a short drive from Texas City



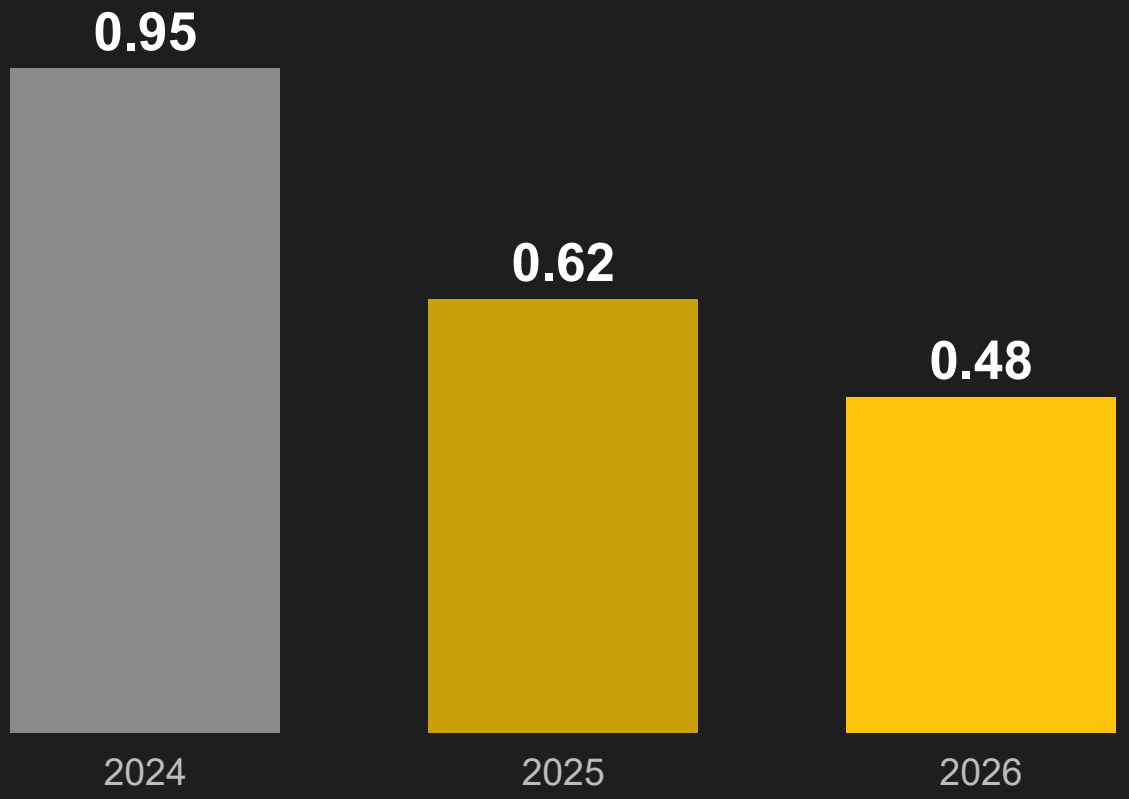

**24/7**
Dispatch and account manager, any hour

**Same day**
Response from Deer Park to Texas City

**6**
TriMax units staged at Deer Park between jobs

# Safety performance

Experience Modification Rate: 1.0 is the industry baseline; lower is better




**A**
ISNetworld safety grade

**0.48**
EMR, 2026

**−49%**
EMR since 2024

**0**
fatalities in 3 years




07 SAFETY & RISK MITIGATION

# Risk mitigation in the field

### JSA & permits

Job-specific JSA reviewed at every shift change; daily toolbox talks

### Isolation first

Valero's isolation and LOTO verified before any connection is made

### Pressure control

Normal 150–300 psi; over-pressure checklist above 500 psi

### Line of fire

Bleed to zero at every pig change; no one in line with closures or hoses

### Hose integrity

Tested & certified hard and flex pipe; whip checks on every connection

### PPE & coil care

FRC and face shields; no galvanized metal on stainless

**Engineered for safety** Retractable cam-actuated stairs, composite work platform, LED lighting

08 CAPABILITIES

# What sets USA DeBusk apart

### Self-performed scope

Pigging, decoking and hydroblasting with our own equipment and crews: one contractor, no subcontractors

### Standardized fleet

6 identical triple-pass TriMax units, 18 engines, with a spare pump on board even on a 2-pass heater

### Engineered pig selection

Decoking pigs matched to each coil and each coke condition

### Data-driven, shared

Durations built from recorded actuals; lessons learned reviewed with Valero after every job



# Trusted across the Valero system




**14** heaters cleaned at 5 Valero refineries

**Texas City, TX**
Coker heater

**Port Arthur, TX**
H-102A, H-102B, F-2000

**Three Rivers, TX**
100-H-1, 100-H-2, H-1102, H-1105, 440-H1

**Meraux, LA**
VU-H-001, VU-H-002

**Wilmington, CA**
20H200, 20H2000, 31H3000

*Coker, vacuum, crude charge and reboiler service*

# Valero case study & lessons learned

## Valero Port Arthur, TX: H-102A & H-102B

Vacuum heaters, March–April 2026

| 16 | 18,496' | 434 |
|---|---|---|
| passes, 2 heaters | coil footage | pigs run |

**Scope** Both 8-pass vacuum heaters pigged three passes at a time, with filtration and smart pig inspection of every pass.

**Result** Every pass signed clean and cleared by smart pig before rig-out. A replacement TriMax was rigged in mid-job within 2 hours, with no break in pigging.

## Pass 6B: coke ring found by smart pig

Mixed-diameter vacuum pass, 3" to 8"

| 0.25" | 24 h | 188 psi |
|---|---|---|
| thick coke ring | dedicated pigging | flow-test reduction |

**Challenge** Inspection found a circumferential coke ring at the inlet of a 5" tube, just before the step-up to 8".

**Lesson learned** On mixed-diameter passes, stage pigs for every diameter and plan time at each step-up before the inspection run.

# Why USA DeBusk for Texas City

## Proven at Valero
14 heaters at 5 Valero refineries, your coker heater among them

## Local
Fleet based in Deer Park, a short drive from Texas City

## Resilient execution
Triple-pass TriMax: a spare pump on every 2-pass heater

## Verified clean
4-point verification and joint sign-off on every pass

## Safe by the numbers
EMR 0.48, ISNetworld A, zero fatalities in 3 years



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# Thank you

## Questions & discussion

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